

Wi-Fi Series

Data Sheet

USE_8720_DF

Wi-Fi+BLE Module

Version

VERSION	REVISION DATE	REVISOR	REVIEWER	REVISED CONTENTS
1.0	28/10/2025	MB		Initial release

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Features:

General

- Dual-band Wi-Fi 802.11 a/b/g/n 1x1 (2.4 GHz & 5 GHz)
- Bluetooth Low Energy (BLE) 5.0 compliant
- Supports LE Secure Connections and Scatternet
- Supports 3 Master links + 1 Slave link
- Operating Temperature: -40 °C to +105 °C
- Operating Voltage: 3.3 V $\pm$ 10 %

MCU Platform

- Chipset: Realtek RTL8720DF-VT1-CG
- Dual-core architecture:
 - KM4 (ARMv8-M, up to 200 MHz)
 - KM0 (ARMv8-M, up to 20 MHz)
- On-chip memory:
 - KM4 SRAM: 512 KB
 - KM0 SRAM: 64 KB
- Embedded Flash: 4 MB
- DMA controller: 6 channels per core
- Secure Boot supported

Wi-Fi Features

- IEEE 802.11 a/b/g/n, 1x1 MIMO
- Frequency range
 - 2.4 GHz: 2412–2484 MHz
 - 5 GHz: 5180–5825 MHz
- Channel bandwidths: 20 / 40 MHz
- Max data rate: up to MCS7
- Security protocols: WEP, WPA, WPA2, WPA3, WPS
- Authentication: Open, shared key, pair-wise key
- Frame aggregation: A-MSDU / A-MPDU
- Low-power Tx/Rx operation
- Antenna diversity supported

Bluetooth Low Energy

- Bluetooth LE version: 5.0
- Supported data rates: 125 kbps / 500 kbps / 1 Mbps / 2 Mbps
- LE Secure Connections supported
- LE Scatternet and multi-link (3M+1S) support
- Co-existence optimized RF design between Wi-Fi and BLE

Peripheral Interfaces

- UART × 4, up to 6 MHz (configurable for log output)
- I²C × 2, up to 400 kHz (fast mode)
- SPI × 3, master/slave up to 50 MHz
- SDIO × 2, host/device up to 50 MHz
- USB 2.0, HS/FS/LS supported
- PWM × 11, configurable 0–100 % duty cycle
- GPIO × 19, fully programmable
- KM4 and KM0 both have a GDMA controller, each with 6 channels.

Antenna Option

- Integrated on-board PCB antenna (dual-band 2.4 / 5 GHz)

Operating Conditions:

- Operating input voltage: (3.3 ± 10%)V
- Operating ambient temperature: -40°C to 105°C

General Description:

The **USE_8720_DF** is a high-performance dual-band Wi-Fi + Bluetooth 5.0 module designed for IoT and embedded wireless applications.

Built around the Realtek RTL8720DF SoC, the module integrates a dual-core MCU (KM4 and KM0), a dual-band Wi-Fi transceiver, and a Bluetooth LE 5.0 controller within a compact 24×16 mm form factor.

This module enables designers to create intelligent products with both Wi-Fi and BLE connectivity while maintaining low power consumption.

Applications include smart home devices, industrial automation, health monitoring, IoT gateways, and consumer electronics.

For more information reach out to us at info@unisemindia.com.

Pin Definition:

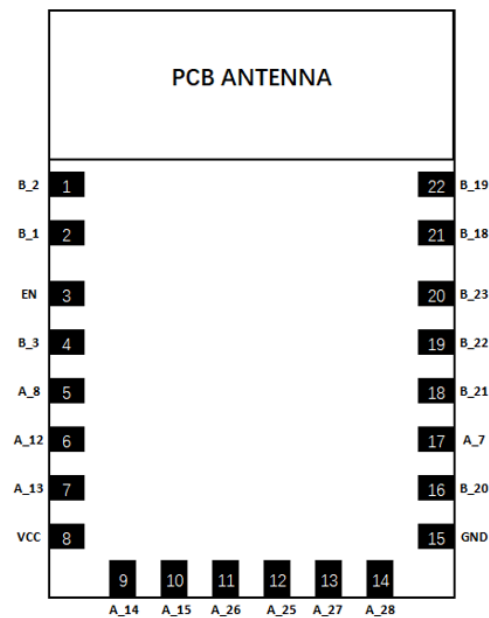


Fig1: Module Pinouts

Pin Descriptions:

Pin No.	Pin Name	Type	Function Description
1	B_2	I/O	GPIOB_2 / UART_RXD
2	B_1	I/O	GPIOB_1 / UART_TXD
3	EN	I	1: Enable chip 0: Shutdown chip
4	B_3	I/O	GPIOB_3 / SWD_CLK
5	A_8	I/O	GPIOA_8 / UART_LOG_RXD
6	A_12	I/O	GPIOA_12 / SPI_MOSI
7	A_13	I/O	GPIOA_13 / SPI_MISO
8	VCC	P	3.3 V Power Supply

9	A_14	I/O	GPIOA_14 / SPI_CLK / UART_RTS
10	A_15	I/O	GPIOA_15 / SPI_CS / UART_CTS
11	A_26	I/O	GPIOA_26 / HSDP
12	A_25	I/O	GPIOA_25 / HSDM
13	A_27	I/O	GPIOA_27 / SWD_DAT
14	A_28	I/O	GPIOA_28 / RREF
15	GND	P	Ground
16	B_20	I/O	GPIOB_20 / SDIO_CMD
17	A_7	I/O	GPIOA_7 / UART_LOG_TXD
18	B_21	I/O	GPIOB_21 / SDIO_CLK
19	B_22	I/O	GPIOB_22 / SDIO_D0
20	B_23	I/O	GPIOB_23 / SDIO_D1
21	B_18	I/O	GPIOB_18 / SDIO_D2
22	B_19	I/O	GPIOB_19 / SDIO_D3

The signal type codes are used in the table:

- I:Input
- O:Output
- P:Power

Strapping Pins:

Pin Name	Pin No.	Default State	Description
A_7	17	Pull-up	1: Normal Mode (default) 0: Flash Download Mode
A_27	13	Pull-up	1: Normal Mode (default) 0: Test Mode

Power Supply DC Characteristics:

Parameter	Symbol	Min	Typ.	Max	Units
Power supply voltage	VCC	3.0	3.3	3.6	V

RF Characteristic

1. Wi-Fi Radio Standard

Wi-Fi Wireless Standard	Description
Wi-Fi frequency range	• 2412 MHz ~ 2484 MHz (2.4 GHz ISM Band) • 5180 MHz ~ 5825 MHz (5 GHz)
Wi-Fi wireless standard	IEEE 802.11 a/b/g/n
Wi-Fi wireless standard modulation	DSSS, DBPSK, DQPSK, CCK and OFDM (BPSK/QPSK/16-QAM/64-QAM)
Wi-Fi wireless data rate	• 802.11a: 6/9/12/18/24/36/48/54 Mbps • 802.11b: 1/2/5.5/11 Mbps • 802.11g: 6/9/12/18/24/36/48/54 Mbps • 802.11n: HT20 MCS0-7, HT40 MCS0-7

1.1 Wi-Fi 2.4 GHz Band RF Transmitter Specification

Parameter	Condition	Min.	Typ.	Max.	Unit
Frequency Range	–	2412	–	2484	MHz
Tx power at the antenna port for the highest power level (25 °C)	1 Mbps DSSS	–	20	–	dBm
	11 Mbps DSSS	–	18	–	dBm
	6 Mbps OFDM	–	19	–	dBm
	54 Mbps OFDM	–	17	–	dBm
	HT20 MCS0	–	18	–	dBm
	HT20 MCS7	–	16	–	dBm
	HT40 MCS0	–	18	–	dBm

	HT40 MCS7	–	16	–	dBm
Tx EVM	1 Mbps DSSS	–	8	–	%
	11 Mbps DSSS	–	8	–	%
	6 Mbps OFDM	–	–5	–	dB
	54 Mbps OFDM	–	–25	–	dB
	HT20 MCS0	–	–5	–	dB
	HT20 MCS7	–	–28	–	dB
	HT40 MCS0	–	–5	–	dB
	HT40 MCS7	–	–28	–	dB
Carrier Suppression	–	–	–	–30	dBc
Harmonic Output Power	2 nd Harmonic	–	–	–45	dBm/MHz
	3 rd Harmonic	–	–	–45	dBm/MHz

1.2 Wi-Fi 2.4 GHz Band RF Receiver Specification

Parameter	Condition	Min.	Typ.	Max.	Unit
Frequency Range	–	2412	–	2484	MHz
802.11b Rx Sensitivity (8 % PER)	1 Mbps DSSS	–	–96	–	dBm
	2 Mbps DSSS	–	–94	–	dBm
	5.5 Mbps DSSS	–	–92	–	dBm
	11 Mbps DSSS	–	–89	–	dBm
802.11g Rx Sensitivity (10 % PER)	6 Mbps OFDM	–	–93	–	dBm
	9 Mbps OFDM	–	–92	–	dBm
	12 Mbps OFDM	–	–91	–	dBm
	18 Mbps OFDM	–	–88	–	dBm
	24 Mbps OFDM	–	–85	–	dBm
	36 Mbps OFDM	–	–82	–	dBm

	48 Mbps OFDM	–	–77	–	dBm
	54 Mbps OFDM	–	–75	–	dBm
802.11n Rx Sensitivity (10 % PER)	HT20 MCS0	–	–93	–	dBm
	HT20 MCS1	–	–90	–	dBm
	HT20 MCS2	–	–87	–	dBm
	HT20 MCS3	–	–84	–	dBm
	HT20 MCS4	–	–81	–	dBm
	HT20 MCS5	–	–76	–	dBm
	HT20 MCS6	–	–75	–	dBm
	HT20 MCS7	–	–73	–	dBm
	HT40 MCS0	–	–91	–	dBm
	HT40 MCS1	–	–87	–	dBm
	HT40 MCS2	–	–84	–	dBm
	HT40 MCS3	–	–81	–	dBm
	HT40 MCS4	–	–78	–	dBm
	HT40 MCS5	–	–73	–	dBm
	HT40 MCS6	–	–72	–	dBm
	HT40 MCS7	–	–70	–	dBm
Maximum Receive Level	1 Mbps DSSS	–	0	–	dBm
	11 Mbps DSSS	–	0	–	dBm
	6 Mbps OFDM	–	0	–	dBm
	54 Mbps OFDM	–	0	–	dBm
	MCS 0	–	0	–	dBm
	MCS 7	–	0	–	dBm

1.3 Wi-Fi 5 GHz Band RF Transmitter Specification

Parameter	Condition	Min.	Typ.	Max.	Unit
Frequency Range	–	5180	–	5825	MHz
Tx power at the antenna port for the highest power level (25 °C)	6 Mbps OFDM	–	17	–	dBm
	54 Mbps OFDM	–	13	–	dBm
	HT20 MCS0	–	15	–	dBm
	HT20 MCS7	–	12	–	dBm
	HT40 MCS0	–	15	–	dBm
	HT40 MCS7	–	12	–	dBm
Tx EVM	6 Mbps OFDM	–	–5	–	dB
	54 Mbps OFDM	–	–25	–	dB
	HT20 MCS0	–	–5	–	dB
	HT20 MCS7	–	–28	–	dB
	HT40 MCS0	–	–5	–	dB
	HT40 MCS7	–	–28	–	dB
Carrier Suppression	–	–	–	–30	dBc

1.4 Wi-Fi 5 GHz Band RF Receiver Specification

Parameter	Condition	Min.	Typ.	Max.	Unit
Frequency Range	–	5180	–	5825	MHz
802.11g Rx Sensitivity (10 % PER)	6 Mbps OFDM	–	–91	–	dBm
	9 Mbps OFDM	–	–91	–	dBm
	12 Mbps OFDM	–	–90	–	dBm
	18 Mbps OFDM	–	–87	–	dBm
	24 Mbps OFDM	–	–84	–	dBm
	36 Mbps OFDM	–	–81	–	dBm

	48 Mbps OFDM	–	–76	–	dBm
	54 Mbps OFDM	–	–74	–	dBm
802.11n Rx Sensitivity (10 % PER)	HT20 MCS0	–	–91	–	dBm
	HT20 MCS1	–	–89	–	dBm
	HT20 MCS2	–	–86	–	dBm
	HT20 MCS3	–	–83	–	dBm
	HT20 MCS4	–	–80	–	dBm
	HT20 MCS5	–	–75	–	dBm
	HT20 MCS6	–	–73	–	dBm
	HT20 MCS7	–	–72	–	dBm
	HT40 MCS0	–	–89	–	dBm
	HT40 MCS1	–	–86	–	dBm
	HT40 MCS2	–	–83	–	dBm
	HT40 MCS3	–	–80	–	dBm
	HT40 MCS4	–	–77	–	dBm
	HT40 MCS5	–	–72	–	dBm
	HT40 MCS6	–	–71	–	dBm
	HT40 MCS7	–	–69	–	dBm
Maximum Receive Level	6 Mbps OFDM	–	0	–	dBm
	54 Mbps OFDM	–	0	–	dBm
	MCS 0	–	0	–	dBm
	MCS 7	–	0	–	dBm

2. Bluetooth LE Radio Standard

2.1 Bluetooth LE RF Transmitter Specification

Parameter	Condition	Min.	Typ.	Max.	Unit
Frequency Range	–	2402	–	2480	MHz
Tx Output Power	LE1M	–10	4.5	10	dBm
	LE2M				
Modulation Characteristics (LE1M)	$\Delta F1$ Avg.	225	–	275	kHz
	$\Delta F2$ Max.	185	–	–	kHz
	Modulation Index ($\Delta F2$ Avg./ $\Delta F1$ Avg.)	0.8	–	–	–
Modulation Characteristics (LE2M)	$\Delta F1$ Avg.	450	–	550	kHz
	$\Delta F2$ Max.	370	–	–	kHz
	Modulation Index ($\Delta F2$ Avg./ $\Delta F1$ Avg.)	0.8	–	–	–
Modulation Characteristics Stable Modulation (LE1M)	$\Delta F1$ Avg.	247.5	–	252.5	kHz
	$\Delta F2$ Max.	185	–	–	kHz
	Modulation Index ($\Delta F2$ Avg./ $\Delta F1$ Avg.)	0.8	–	–	–
Modulation Characteristics Stable Modulation (LE2M)	$\Delta F1$ Avg.	495	–	505	kHz
	$\Delta F2$ Max.	370	–	–	kHz
	Modulation Index ($\Delta F2$ Avg./ $\Delta F1$ Avg.)	0.8	–	–	–

2.2 Bluetooth LE RF Receiver Specification

Parameter	Condition	Min.	Typ.	Max.	Unit
Frequency Range	–	2402	–	2480	MHz
Rx Sensitivity @ 30.8 % PER	LE1M	–	–99	–	dBm
	LE2M	–	–95	–	dBm

Module Electrical Characteristics:

1. Module Operating Conditions

Symbol	Parameter	Min.	Typ.	Max.	Units
VCC	Power supply voltage	3.0	3.3	3.6	V
Ta	Ambient operating temperature	−40	–	105	°C
Ts	Storage temperature	−40	–	125	°C

2. Module DC Characteristics

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Units
VIH	Input-High Voltage	LVTTL	2.0	–	–	V
VIL	Input-Low Voltage	LVTTL	–	–	0.8	V
VOH	Output-High Voltage	LVTTL	2.4	–	–	V
VOL	Output-Low Voltage	LVTTL	–	–	0.4	V
VT+	Schmitt-trigger High Level	–	1.78	1.87	1.97	V
VT–	Schmitt-trigger Low Level	–	1.36	1.45	1.56	V
IIL	Input-Leakage Current	VIN = 3.3 V or 0	−10	±1	10	μA

Module Block Diagram:

This module includes the chipset, crystal component, R/L/C components for RF matching, decoupling and RF radio antenna.

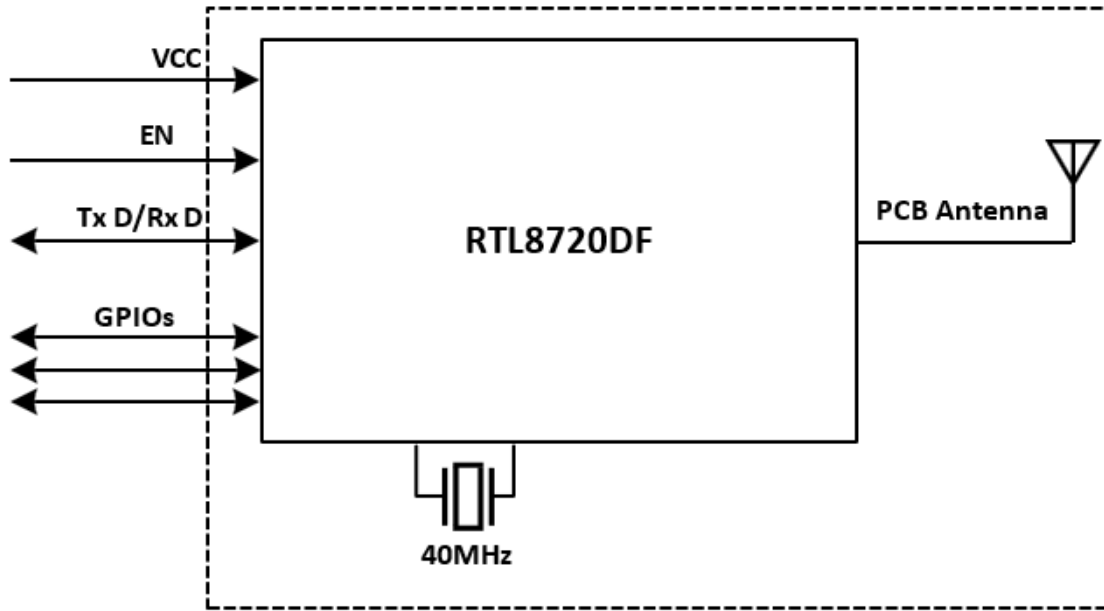


Fig2: Block Diagram

Module Reference Schematics :

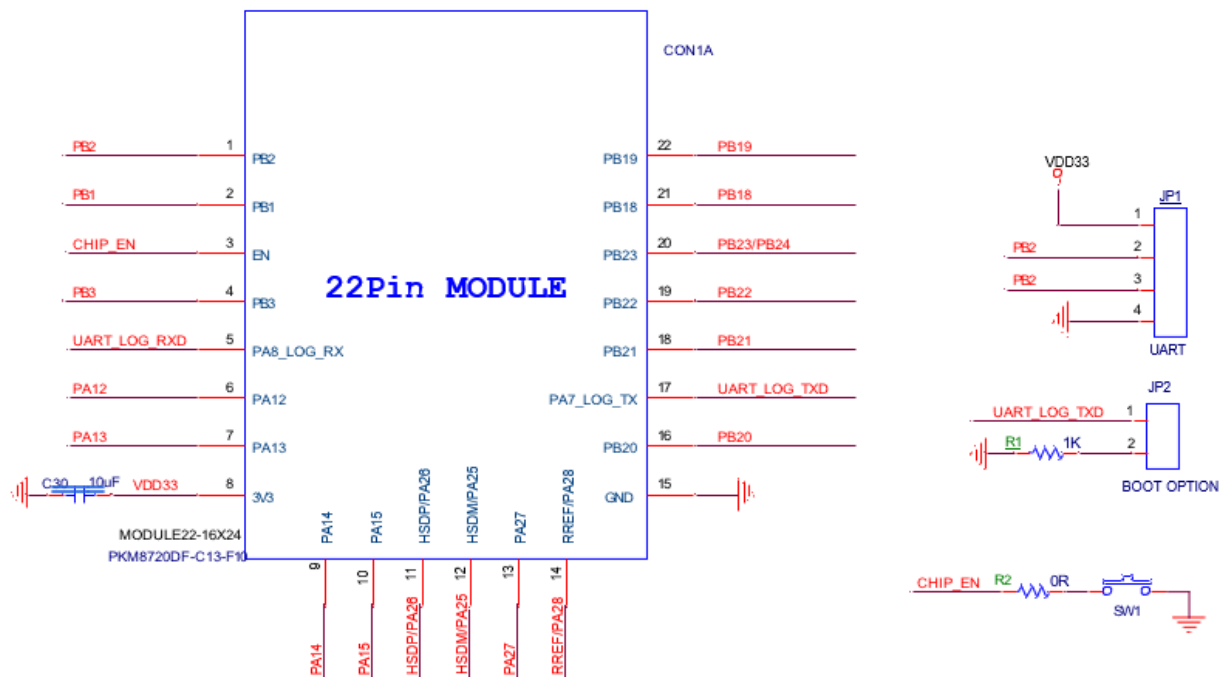


Fig 3: Module Reference schematics

Physical Dimensions :

Module dimension: $24 \pm 0.2\text{mm}$ (L) x $16 \pm 0.2\text{mm}$ (W) x $2.3 \pm 0.1\text{mm}$ (H).

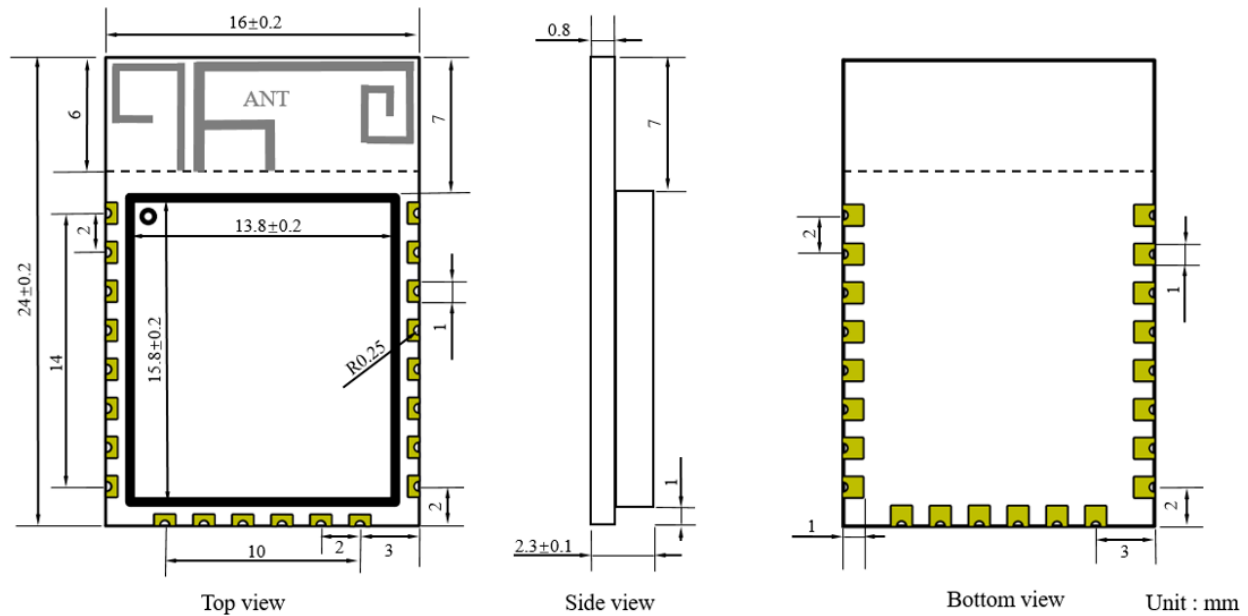


Fig4 : Module Physical Dimensions

Recommended Assembly:

Signal trace and power supply trace, don't cross layout, as shown in below Fig. To avoid crosstalk, affect the receiving sensitivity.

Assembly recommendation

Underneath the module antenna and RF circuit on the main board PCB copper needs to be cleared, and placed close to the main board edge, as shown in below Fig.

For smaller coin cell batteries, with high worst-case end-of-life equivalent source resistance, a 22- μF VDDS input capacitor must be used to ensure compliance with this slew rate. The recommended one is to use MLCC for low ESR.

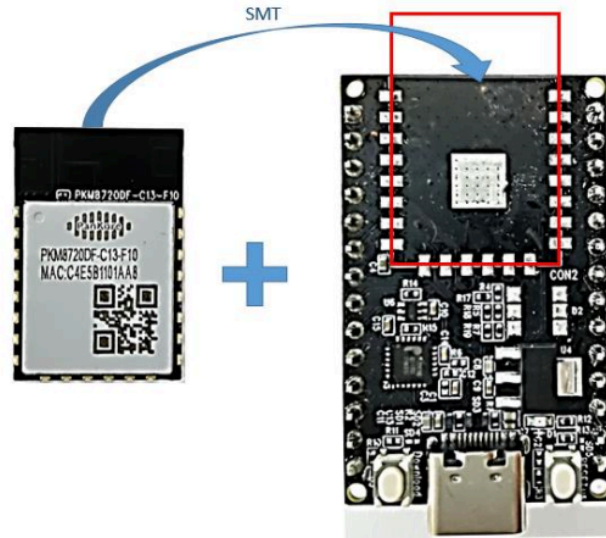


Fig4 : Module Assembly 3D

The module assembled in the motherboard, must pay attention to: Copper around the antenna and on all layers must be cleared. The module antenna should be on the edge, and there is no metal or obstruct electromagnetic radiation material to surround the antenna area.

Module placement strategy:

BLE/Sub-1GHz modules are RF devices that require proper placement on PCB to ensure optimal performance. The antenna on the PCB has (about) an omnidirectional radiation pattern. To maximize antenna efficiency, an adequate grounding plane must be provided under the module. However, the areas underneath and surrounding the antenna area must be free of copper. The position of the module on the host board and overall design of the product enclosure contribute to antenna performance. Poor design affects radiation patterns and can result in reflection, diffraction, and/or scattering of the transmitted signal thus limiting the range.

Basic guidelines:

- ❖ Never place the ground plane or route copper traces directly underneath the antenna portion of the module.
- ❖ Never place the antenna close to metallic objects.
- ❖ Keep wiring, components and objects away from the antenna.
- ❖ Do not place the antenna in a metallic or metalized plastic enclosure.
- ❖ Enclosure walls should be 1cm or more away from the antenna in all directions.
- ❖ If possible, mount the antenna overhanging the edge of the host board (Motherboard). Add an uninterrupted ground plane on the host board, directly underneath the top shielded zone of the module, up to the PCB edge. Adding a ground plane will allow traces to be run on the bottom side of the host board if required.

- ❖ If the antenna cannot be mounted in an overhanging position, then provisions must be made to keep the area clear of copper.

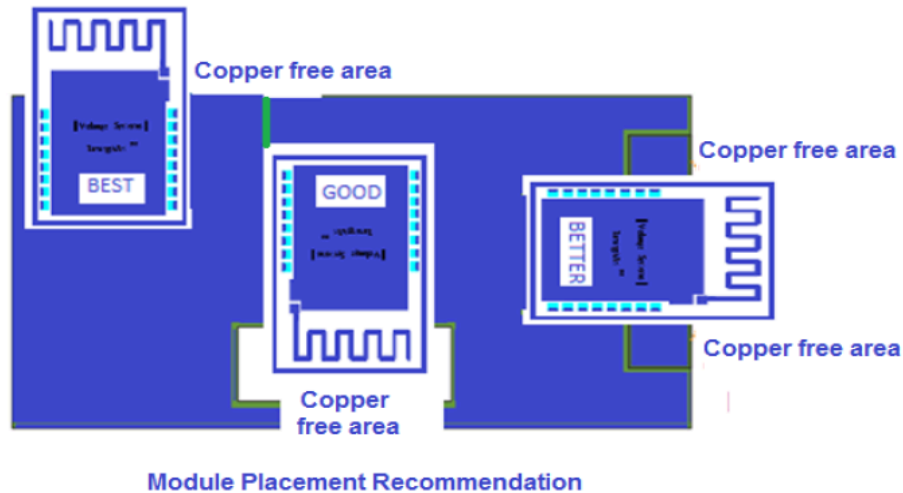
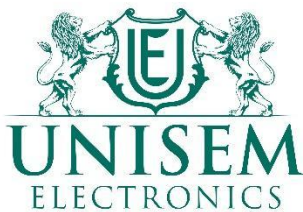


Fig 5

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